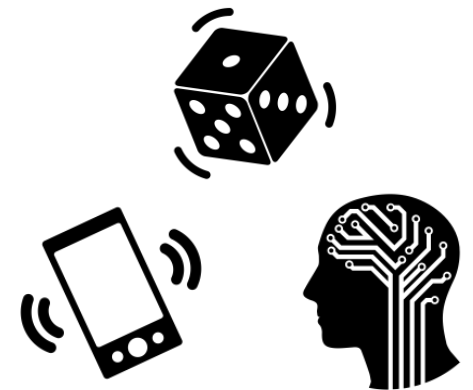


Probability & Random Variables

2026 Spring

Prof. Bang Chul Jung



Lecture 09

Asymptotic Behaviors of Random Variables

Target:

- To learn the **two** most remarkable findings in probability theory.
 - Law of large numbers & Central limit theorem



Roadmap

- (1) Weak Law of Large Numbers: Result and Meaning
- (2) Central Limit Theorem: Result and Meaning

Our Interest: **Sum** of Random Variables

- **Example 1.** n students who decide their presence, depending on their feeling. Each student is happy or sad at random, and only happy students will show their presence. How many students will show their presence?
- **Example 2.** I am hearing some sound. There are n noisy sources from outside.
- $X_1, X_2, \dots, X_n$: i.i.d (independent and identically distributed) random variables
- $\mathbb{E}[X_i] = \mu, \text{var}[X_i] = \sigma^2$
- Our interest is to understand how the following sum behaves:

$$S_n = X_1 + X_2 + \dots + X_n$$

What About Distribution of S_n ?

$$S_n = X_1 + X_2 + \dots + X_n$$

- Figure out the distribution of S_n . Very challenging. Even just for $Z = X + Y$, finding the distribution, for example, requires the complex **convolution**.

$$p_Z(z) = \mathbb{P}(X + Y = z) = \sum_x p_X(x)p_Y(z - x)$$

- Easy case: Sum of normal rvs = a normal rv, however, generally very challenging.
- **Possible approach**. Take a certain **scaling** with respect to n that corresponds to a **new glass**, and investigate the system for large n

Sample Mean

- Consider the **sample mean**, and try to understand how S_n behaves:

$$M_n = \frac{S_n}{n} = \frac{X_1 + X_2 + \dots + X_n}{n}$$

- $\mathbb{E}(M_n) = \mu$, $\text{var}(M_n) = \sigma^2/n$
- For large n , the variance $\text{var}(M_n)$ decays. We expect that, for large n , M_n loses its randomness and **concentrates around μ** .
- We call this **law of large numbers (LLN)**.

Weak Law of Large Numbers (WLLN)

$$M_n = \frac{S_n}{n} = \frac{X_1 + X_2 + \dots + X_n}{n}$$

- Roughly, M_n concentrates around μ

Weak law of large numbers

M_n converges to μ in probability, i.e., $M_n \xrightarrow{\text{in prob.}} \mu$

- Why “Weak”? There exists a stronger version, which we call “strong” law of large numbers. We will not cover the strong law of large numbers in this class.
- If we take the scaling of S_n by $1/n$, it behaves like a **deterministic** number. This significantly simplifies how we understand the world.
- For example, assume that a large number of identically distributed noises come to you. Then, you can roughly approximate it as **($n \times$ average noise)**

Roadmap

(1) Weak Law of Large Numbers: Result and Meaning

(2) Central Limit Theorem: Result and Meaning

Central Limit Theorem: Formalism

- $S_n = X_1 + X_2 + \cdots + X_n,$ $Z_n = \frac{S_n - n\mu}{\sigma\sqrt{n}}$

Central Limit Theorem

Z_n converges to Z in distribution, where $Z \sim \mathcal{N}(0, 1)$.

- Very surprising!
- Irrespective of the distribution of X_i , Z is normal.

LLN vs. CLT: Different Scaling Glasses

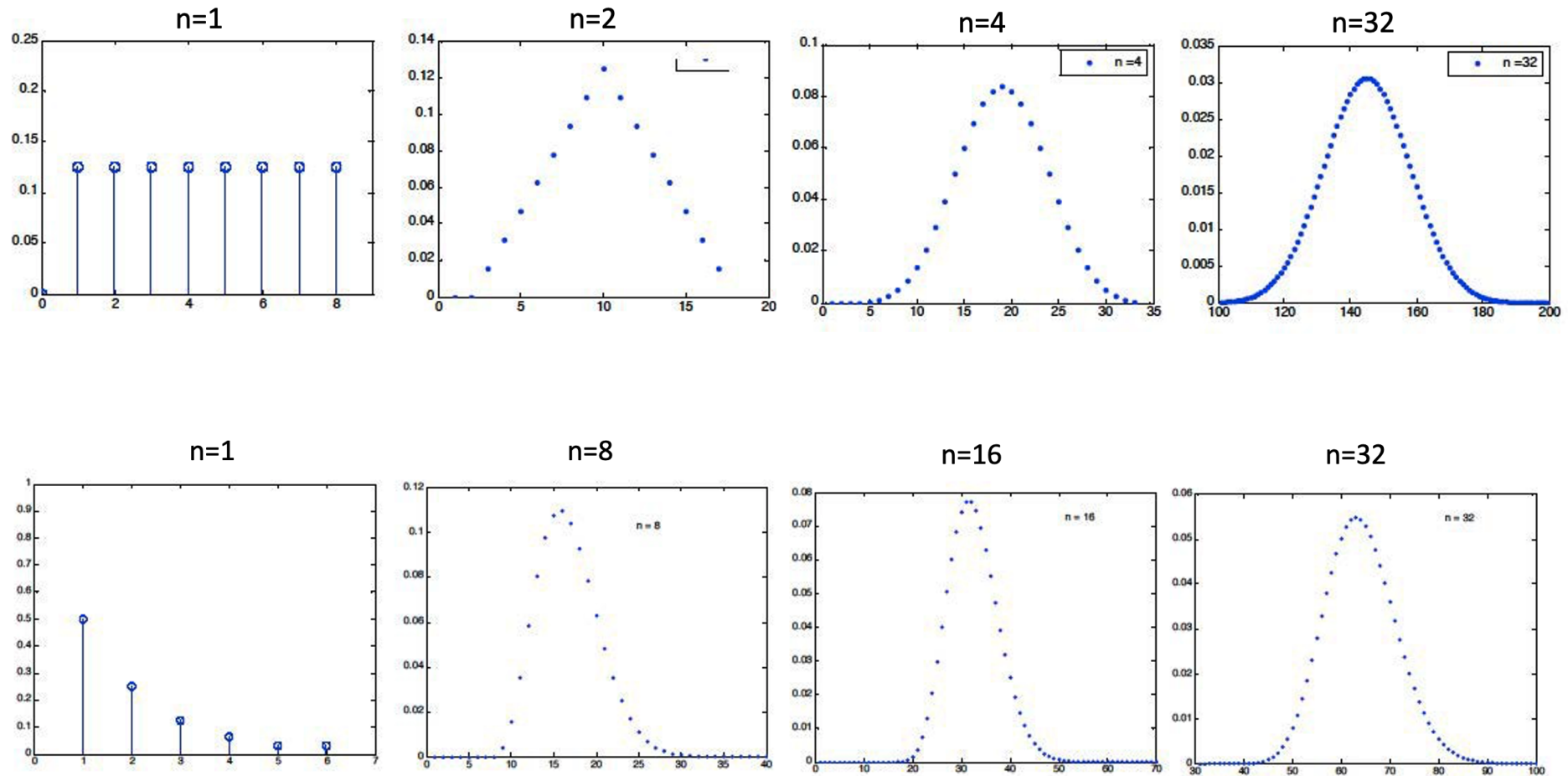
- For simplicity, assume that $\mathbb{E}(X_i) = 0$ and $\text{var}(X_i) = 1$, $i = 1, 2, \dots, n$
- Law of Large Numbers

Scaling S_n by $1/n$, you go to a **deterministic** world.

- Central Limit Theorem

Scaling S_n by $1/\sqrt{n}$, you still stay at the **random** world, but not an arbitrary random world. That's the **normal** random world, not depending on the distribution of each X_i .

CLT: Examples of Required n



***Thank You for Your
Attention!***

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